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**Welcome
To
Network for you
CISCO NetFlow**



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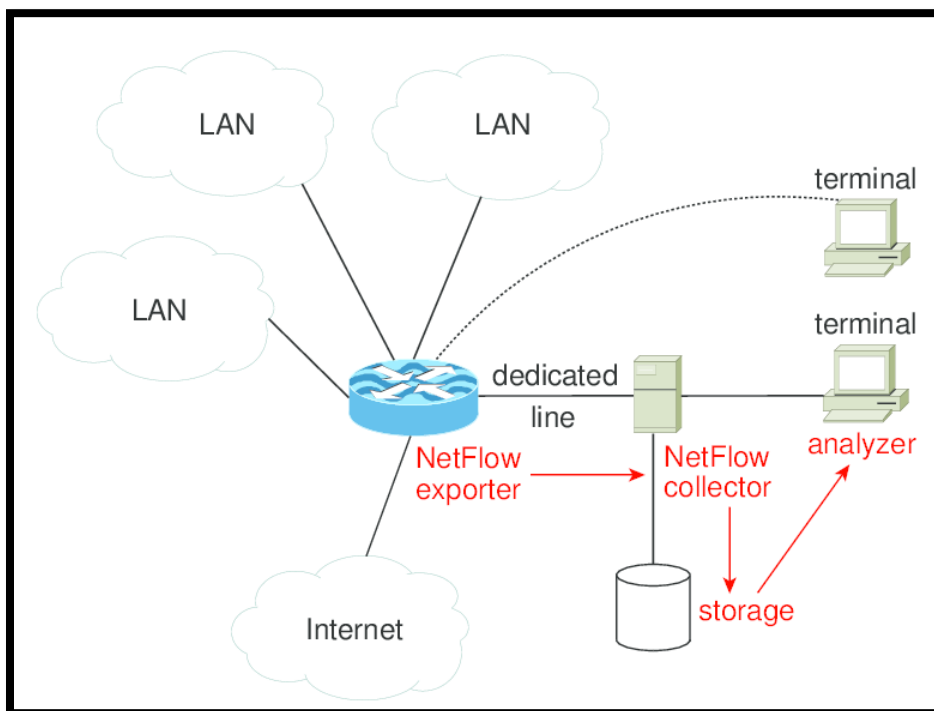
CISCO NetFlow:

- **Network management protocols like SNMP allow us to monitor our network.**
- We can check things like **cpu load, memory usage, interface status and even the load of an interface**
- One of the things we can't do with those tools (SNMP) is **tracking all flows in our network.**
- A flow is a stream of packets that share the same characteristics like source/destination port, source/destination address, protocol, type, service marking, etc.
- **NetFlow allows us to track these flows on our network.**
- We can use this information to solve problems like bottlenecks, **identify what applications** are used, how much bandwidth they use etc.
- For each of the flows, NetFlow will track the number of packets sent, bytes sent, packet sizes and more.
- We can configure your router to keep track of all flows and then export them to a central server where we analyze our traffic.
- NetFlow is an application for collecting IP traffic information.
- NetFlow is a protocol developed by Cisco Systems to record all IP traffic flows.
- Flow characteristics are like source, destination port, address, protocol, type etc.
- NetFlow allows tracking these flows on in the network.
- NetFlow track the number of packets sent, bytes sent, packet sizes and more.
- Configure router to keep track of all flows and then export them to central server.
- NetFlow is a great protocol to get an insight in the network traffic.
- NetFlow allows seeing real time data on who/what is eating the bandwidth.
- NetFlow is used to collect data flows from routers & switches interfaces.
- NetFlow is an application that provides statistics on packets flowing through routers.
- NetFlow captures statistics on IP flows through a device.
- NetFlow allows collecting traffic and analyzing it through a program.
- NetFlow is configured in the interface configuration mode on a router.
- NetFlow monitor of ingress traffic, egress traffic, or both ingress or egress traffic.
- Specify the IP address of the NetFlow collector & UDP port of collector is listening.
- Provides statistics on packets flowing through a router or a switch.
- NetFlow collect and export the data to enable network & security monitoring.
- NetFlow collect & export data for network planning, traffic analysis & IP accounting.
- NetFlow records are exported to a NetFlow collector using UDP.
- The standard value is UDP port 2055, but other values can be set 9555 or 9995.
- NetFlow can be used for network accounting and security auditing.
- NetFlow consumes additional memory of devices to process.



Or in Simple words we can say

- One of the most important tasks of a network administrator is to monitor the health of our networks, learn how our bandwidth is being used, what applications are consuming it, when it needs upgrade...
- Although monitoring protocols like SNMP and SPAN (port mirroring) can help us answer some questions but they are not enough to give us an insightful view of our networks.
- Luckily we have another amazing tool: NetFlow!
- NetFlow is a networking analysis protocol that gives the ability to collect detailed information about network traffic as it flows through a router interface.
- NetFlow helps network administrators answers the questions of who (users), what (application), when (time of day), where (source and destination IP addresses) and how network traffic is flowing.





NetFlow Versions:

Version 1:

- First implementation, now obsolete, and restricted to IPV4 only.

Versions 2:

- Cisco internal version never released.

Version 3:

- Cisco internal version never released.

Version 4:

- Cisco internal version never released.

Version 6:

No longer supported by Cisco Encapsulation information.

Version 6 is not compatible with Cisco routers.

Version 7:

- Cisco-specific version for Catalyst 5000 series switches.
- Version 7 is not compatible with Cisco routers.

Version 8:

- Choice of aggregation schemes in order to reduce resource usage.

Version 5:

- Fixed format that cannot be added or extended.
- NetFlow version 5 only support IPv4.
- NetFlow version 5 added BGP support.
- Export data from main cache only.
- No real concept of ingress & egress flows.
- NetFlow added flow sequence numbers & additional fields.
- NetFlow Version 5 is standard & most common NetFlow version.

Version 9:

- NetFlow Version 9 support IPV4 and IPv6.



- Not backwards compatible with any previous version.
- Added additional information to flows & template based.
- Exports data from main & aggregation cache.
- NetFlow Version 9 support for MPLS.
- Support flow-record format known as Flexible NetFlow technology.
- NetFlow Version 9 is most important NetFlow version.

NetFlow uses the following fields to identify a unique flow:

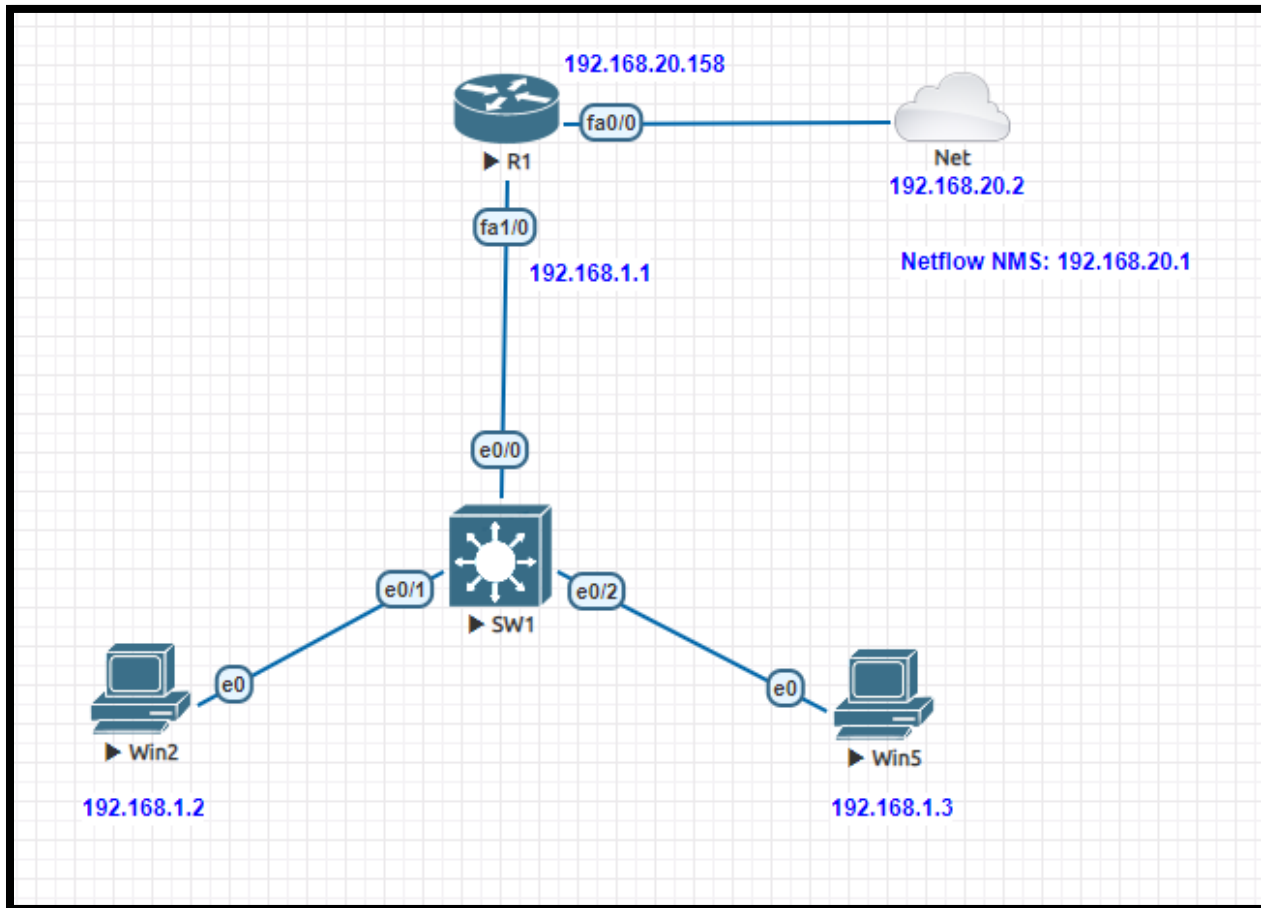
- Source IP address
- Source port number
- Destination IP address
- Destination port number
- Layer 3 Protocol Type
- Type of service
- Logical input interface

Basic configuration:

Netflow Configuration Command	Description
R1(config)#ip flow-export source f0/0	We have to do is tell the router on what interfaces to track the flows
R1(config)#int f0/0 R1(config-if)#ip flow ingress R1(config-if)#ip route-cache flow R1(config-if)# ip flow egress R1(config-if)# exit	We will use the ip route-cache flow command. When you use this command, it will track all flows on the physical and all sub-interfaces. You can also use the ip flow egress or ip flow ingress commands
R1(config)#ip flow-export destination 192.168.20.1 2055	The router will export all flows to 192.168.20.1 with destination UDP port 2055
R1(config)#ip flow-export version 5	I will configure the router to use version 5
R1(config)#ip flow-cache timeout active 1	Export flow records every minute.
R1(config)#ip flow-cache timeout inactive 10	
Show Commands	Sh ip flow export
	Sh ip cache flow



Lab time:



R1 Configuration:

```
en
config t
hostname R1

int f0/0
ip add dhcp
```

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```
no sh
```

```
int f1/0
```

```
ip add 192.168.1.1 255.255.255.0
```

```
no sh
```

```
ip domain-lookup
```

```
ip name-server 8.8.8.8
```

Netflow Configuration:

```
R1(config)#int f0/0
```

```
R1(config-if)#ip flow ingress
```

```
R1(config-if)#ip route-cache flow //I will use the ip route-cache flow command. When you use this command, it will track all flows
```

```
on the physical and all sub-interfaces. You can also use the ip flow egress or ip flow ingress commands
```

```
R1(config-if)#exit
```

```
R1(config)#ip flow-export destination 192.168.20.1 2055 //The router will export all flows to 192.168.20.1 with destination UDP port 2055
```

```
R1(config)#ip flow-export source f0/0 // we have to do is tell the router on what interfaces to track the flows
```

```
R1(config)#ip flow-export version 5 //I will configure the router to use version 5
```

```
R1(config)#ip flow-cache timeout active 1 //export flow records every minute.
```

```
R1(config)#ip flow-cache timeout inactive 10
```

Show Command

```
R1#show ip flow export
```

```
R1# show ip cache flow
```

```
R1#sh ip flow export
```

```
Flow export v5 is enabled for main cache
```

```
Export source and destination details :
```

```
VRF ID : Default
```

```
Destination(1) 192.168.20.1 (2055)
```

```
Version 5 flow records
```

```
56 flows exported in 15 udp datagrams
```

```
0 flows failed due to lack of export packet
```

```
0 export packets were sent up to process level
```

```
0 export packets were dropped due to no fib
```

```
0 export packets were dropped due to adjacency issues
```

```
0 export packets were dropped due to fragmentation failures
```

```
0 export packets were dropped due to encapsulation fixup failures
```

```
R1#sh ip cache
```



```
R1#sh ip cache flow
IP packet size distribution (29492 total packets):
 1-32   64   96  128  160  192  224  256  288  320  352  384  416  448  480
 .000 .223 .000 .000 .000 .000 .000 .000 .000 .000 .000 .000 .002 .000 .000

 512  544  576 1024 1536 2048 2560 3072 3584 4096 4608
 .000 .003 .001 .008 .753 .000 .000 .000 .000 .000 .000

IP Flow Switching Cache, 4456704 bytes
 2 active, 65534 inactive, 102 added
2819 age polls, 0 flow alloc failures
Active flows timeout in 1 minutes
Inactive flows timeout in 10 seconds
IP Sub Flow Cache, 533256 bytes
 2 active, 16382 inactive, 57 added, 57 added to flow
 0 alloc failures, 0 force free
 1 chunk, 1 chunk added
last clearing of statistics never
```

Protocol	Total Flows	Flows /Sec	Packets /Flow	Bytes /Pkt	Packets /Sec	Active(Sec) /Flow	Idle(Sec) /Flow
TCP-WWW	48	0.0	550	1137	22.2	29.5	11.0
TCP-other	30	0.0	3	274	0.0	2.0	11.3
UDP-DNS	6	0.0	1	68	0.0	0.5	14.7
UDP-other	15	0.0	2	164	0.0	1.1	14.8
IGMP	1	0.0	5	40	0.0	0.2	15.1
Total:	100	0.0	265	1131	22.3	15.0	11.9

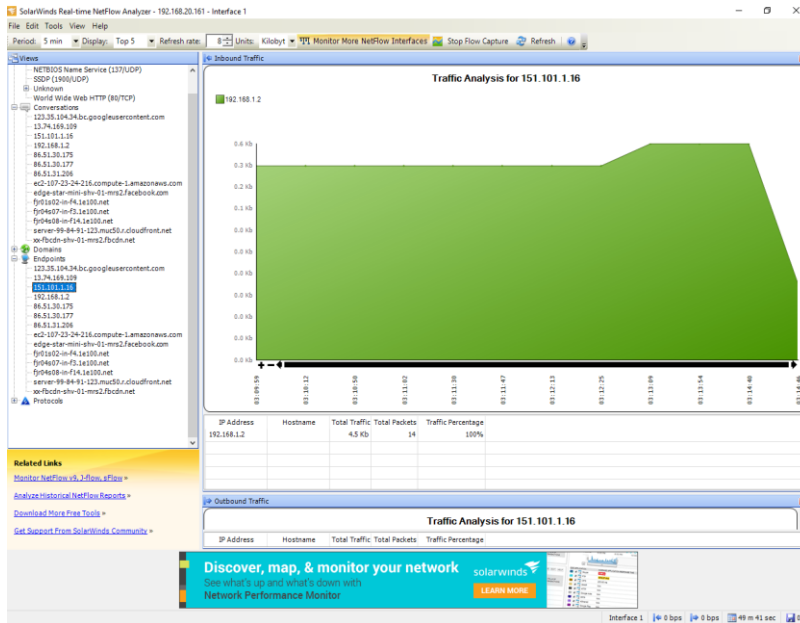
SrcIf	SrcIPAddress	DstIf	DstIPAddress	Pr	SrcP	DstP	Pkts
Fa0/0	34.104.35.123	Fa1/0	192.168.1.2	06	0050	C041	1
Fa1/0	192.168.1.2	Fa0/0*	34.104.35.123	06	C041	0050	1
Fa1/0	192.168.1.2	Fa0/0*	86.51.30.175	06	C04B	0050	744
Fa0/0	86.51.30.175	Fa1/0	192.168.1.2	06	0050	C04B	3163



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For Lab Purpose we will use SolarWinds Real Time Netflow Analyzer you can download from Net free.



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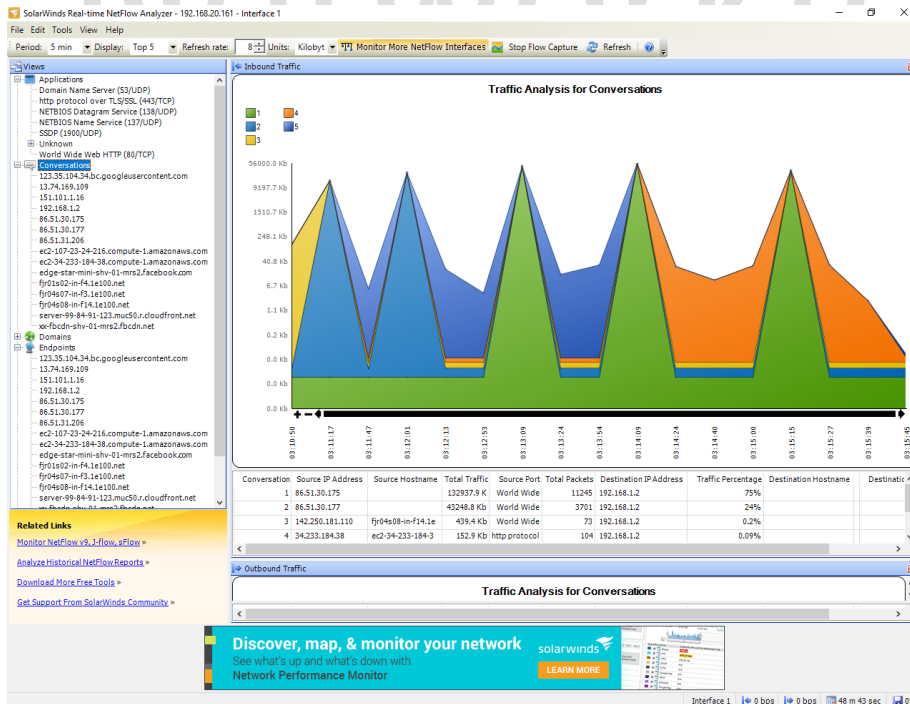
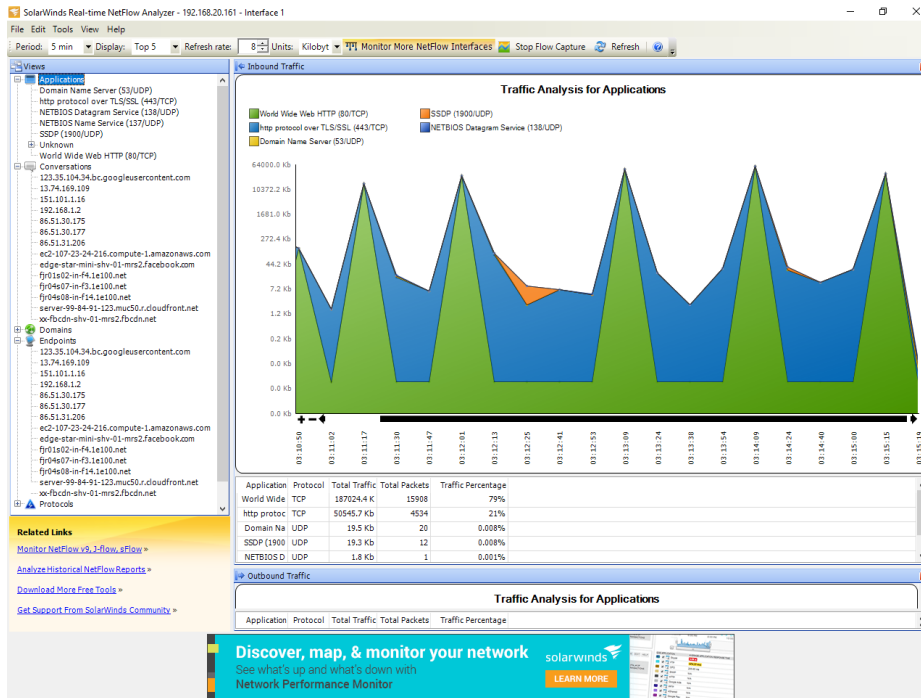
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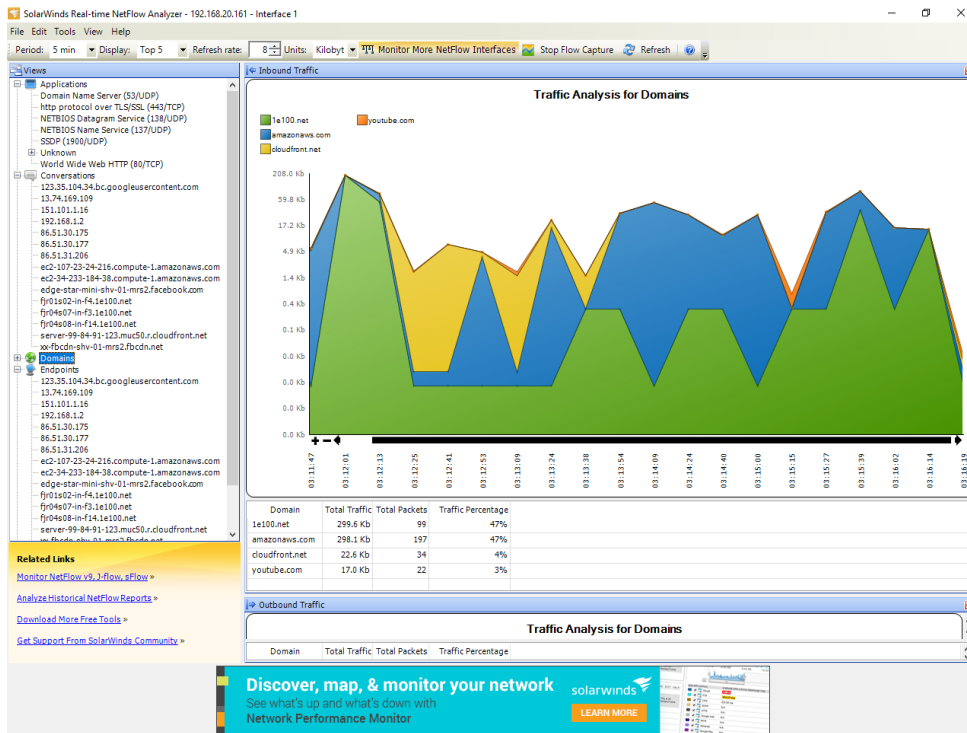
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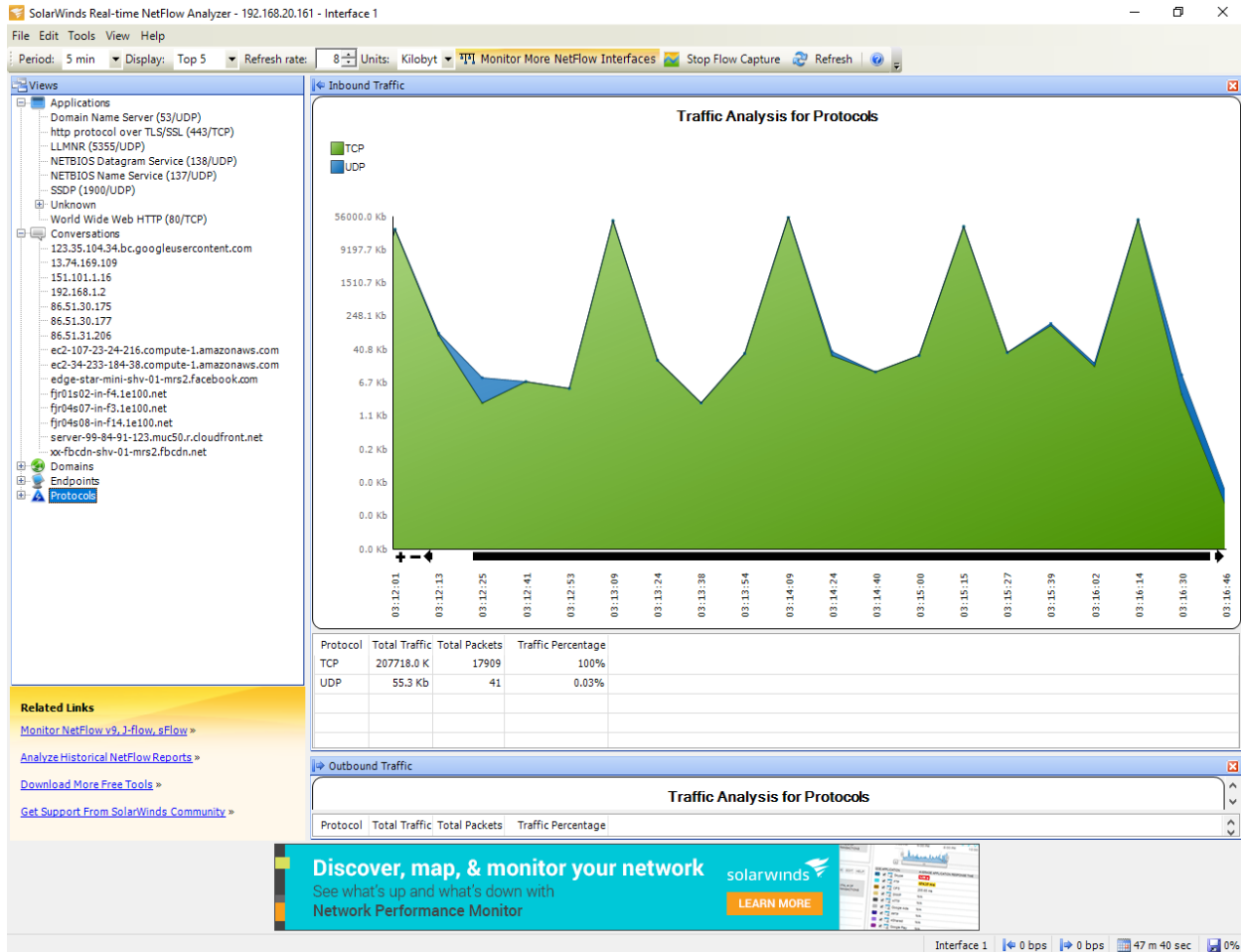
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Top Talker

R1(config)# ip flow-top-talkers

R1(config-flow-top-talkers)# top 10

R1(config-flow-top-talkers)# sort-by packets

Sh Command

R1#sh ip flow top-talkers

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R1#sh ip flow top-talkers

SrcIf	SrcIPAddress	DstIf	DstIPAddress	Pr	SrcP	DstP	Pkts
Fa0/0	86.51.30.175	Fa1/0	192.168.1.2	06	0050	C0BE	1514
Fa0/0	216.58.208.225	Fa1/0	192.168.1.3	06	01BB	C02A	1493
Fa0/0	172.217.21.36	Fa1/0	192.168.1.3	06	01BB	C028	627
Fa1/0	192.168.1.3	Fa0/0*	216.58.208.225	06	C02A	01BB	514
Fa1/0	192.168.1.2	Fa0/0*	86.51.30.175	06	C0BE	0050	366
Fa0/0	157.240.195.35	Fa1/0	192.168.1.3	06	01BB	C023	167
Fa1/0	192.168.1.3	Fa0/0*	172.217.21.36	06	C028	01BB	165
Fa0/0	86.51.94.58	Fa1/0	192.168.1.3	06	0050	C024	86
Fa0/0	157.240.195.35	Fa1/0	192.168.1.3	06	01BB	C032	11
Fa0/0	52.205.139.193	Fa1/0	192.168.1.2	06	01BB	C16E	11

10 of 10 top talkers shown. 28 flows processed.

R1#sh ip flow top-talkers

SrcIf	SrcIPAddress	DstIf	DstIPAddress	Pr	SrcP	DstP	Pkts
Fa0/0	172.217.21.36	Fa1/0	192.168.1.3	06	01BB	C028	1870
Fa1/0	192.168.1.3	Fa0/0*	172.217.21.36	06	C028	01BB	525
Fa0/0	157.240.195.35	Fa1/0	192.168.1.3	06	01BB	C023	457
Fa0/0	172.217.19.3	Fa1/0	192.168.1.3	06	01BB	C02C	219
Fa0/0	86.51.30.175	Fa1/0	192.168.1.2	06	0050	C0BE	42
Fa1/0	192.168.1.3	Fa0/0*	157.240.195.35	06	C023	01BB	32
Fa1/0	192.168.1.3	Fa0/0*	172.217.19.3	06	C02C	01BB	31
Fa0/0	142.250.181.78	Fa1/0	192.168.1.3	06	01BB	C041	25
Fa1/0	192.168.1.3	Fa0/0*	142.250.181.78	06	C041	01BB	14
Fa0/0	52.205.139.193	Fa1/0	192.168.1.2	06	01BB	C16E	14

10 of 10 top talkers shown. 43 flows processed.